

GO WITH THE FLOW: OPTIMAL SAILBOAT RACING

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ABSTRACT. In this paper we will try to model the optimal path in the historical M&M 100 Miler sailing race that takes place in Lake Michigan. Many forces impact optimal sailing, including wind direction, current, and waves, making this a challenging problem to model accurately. The primary questions we work to answer are what the simplest model is that rediscovers real sailing techniques, such as tacking, and how many parameters we can add into our model while still being able to solve it and getting reasonable results. We solve the classic Zermelo Navigation Problem assuming constant boat velocity and fixed current drift vector field. Further iterations involve relying on wind instead of a motor, taking into account the impact of the islands on wind and current speed, and including the impacts of being near Wisconsin on the wind. Our more accurate models were able to find a path that matched the speeds of real-world sailors from the year we got current and wind data.

1. BACKGROUND

The primary question this paper seeks to determine is the simplest model that would re-discover real world sailing techniques, particularly tacking and jibing.

Every breadmaker has noticed that small changes in initial dough conditions can lead to wide variations in the quality of the baked bread. By having a good model for how yeast interacts with the other ingredients in bread, we will be able to better understand the breadmaking process and how to optimize initial conditions.

2. MATHEMATICAL REPRESENTATION

In this section we go over the models we built in detail.

3. SOLUTION

[Eth25]

4. INTERPRETATION

Appropriateness of models.

Potential improvements.

Impact.

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REFERENCES

- [Eth25] EthanPrimes. `modeling_yeast_spread`. https://github.com/EthanPrimes/modeling_yeast_spread, 2025. GitHub repository, MIT license, accessed 2025-12-03.